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DISCORDANT MID-CRETACEOUS PALEOMAGNETIC POLE FROM THE ZAZA TERRANE OF CENTRAL CUBA

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Abstract. Thermal demagnetization of samples from five sites in mid-Cretaceous volcaniclastic and carbonate rocks of south-central Cuba reveals two or more components of magnetization. A pre-folding magnetite borne component is recognized in all, but was difficult to isolate from secondary components in many specimens. Remagnetization circle analysis, using data for 42 specimens from all five sites, yields a pole at 30.7°N Lat., 193.3°E Long. This pole is discordant with respect to the North American APWP, indicating 43°± 16° of anticlockwise rotation and 8°± 6° of northward displacement since the mid-Cretaceous. This result suggests that part or all of Cuba was transported on the Caribbean plate before accretion to North America in the Eocene.

Introduction

Geological and geophysical features of the Greater Antilles (Figure 1A) suggest that the tectonic evolution of the Caribbean-North American plate boundary has occurred in two stages, a pre-Late Eocene period of convergence followed by sinistral transform motion that continues today [e.g. Pindell and Dewey, 1982; Donnelly, 1989]. The present plate boundary is a diffuse zone, as shown by neotectonic studies [e.g. Sykes et al., 1982], and the rigid block rotations in the Greater Antilles inferred from paleomagnetic data have been attributed to distributed shear within this zone [e.g. Duncan and Hargraves, 1984; Gose, 1985].

Correlation of Cuba's Cretaceous island arc(s) with those of Jamaica, Hispañola and Puerto Rico favors an original location of all four islands on the same (Caribbean) plate, with Cuba accreting to North America as the plate boundary jumped south to the Swan-Oriente fracture zone (Figure 1A), roughly synchronous with the onset of spreading at the Cayman Rise [Rosencrantz et al., 1988]. This model would predict little or no post-Eocene rotation of Cuban blocks, because the distributed sinistral fault zones typical south of the Oriente Fault are unrecognized in Cuba.

The Zaza Terrane as defined by Hatten et al. [1989] consists of approximately 6500 m of island arc volcanic and volcaniclastic strata ranging in age from Neocomian(?) to Campanian, deformed into north-vergent folds whose axes generally trend about 270° to 320°. Zaza strata are overlain unconformably by Maastrichtian through Early Eocene epiclastic sedimentary rocks and minor reefal limestones. The nature of the contact between Zaza strata and the structurally underlying ultramafic unit (um in Figure 1B) is uncertain, but is considered herein to be related to the Las Villas thrust whose principal surface juxtaposes the ultramafics against the Las Villas Terrane [Hatten et al., 1989].

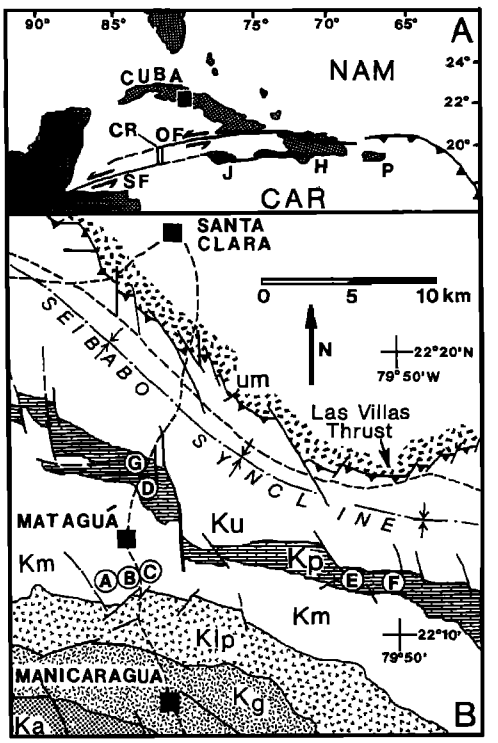


Fig. 1. (A) Tectonic features of the northern Caribbean plate margin. SF, Swan fracture zone; OF, Oriente fracture zone; CR, Cayman Rise spreading center; J, Jamaica; H, Hispañola; P, Puerto Rico; CAR and NAM denote Caribbean and North American plates, respectively. Black square in central Cuba is the area shown in figure 1B. (B) Geology of the sampling region, with sample sites indicated by circled letters (Table 1). Dashed lines are paved roads. Map units are: Ka, Mabujina amphibolite complex; Kg, Manicaragua granodiorite; Klp, Los Pasos Fm.; Km, Mataguá Fm.; Kp, Provinciál Fm.; Ku, undifferentiated Upper Cretaceous strata; um, ultramafic rocks.

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Samples were collected from the Mataguá (three sites) and Provinciál (four sites) formations on the southern limb of the Seibabo syncline (Figure 1B). The Mataguá Formation comprises andesitic tuffs and tuffaceous sandstones of Aptian-Albian age [Diaz de Villalvilla, 1985] (P. R. Renne, unpublished $^{40}Ar/^{39}Ar$ data,1991). The overlying Provinciál Formation is composed of limestones and calcareous tuffs of latest Albian to Cenomanian age [Hatten et al., 1989; Diaz de Villalvilla, 1985]. Rocks of the Mataguá Formation have experienced low-grade burial metamorphism; secondary quartz ± prehnite ± albite ± chlorite ± calcite indicate maximum temperatures of 200 to 300°C.

Paleomagnetic Results

Nine to 16 oriented cores from each of seven sites were oriented by magnetic compass and clinometer. Each sample was thermally demagnetized in 16 to 22 steps. Provinciál samples were measured with a cryogenic magnetometer; Mataguá samples with a spinner.

Mataguá Formation

The Mataguá samples (NRM 10^{-1} to 10^{-2} A/m) revealed two or more components of magnetization (Figure 2A). A low temperature component (A), below 250°C parallels the Quaternary field. Component (B) has sharp unblocking temperatures (560 to 590°C), and is westerly with a shallow inclination. Progressive demagnetization produces great circle trajectories of directions representing mixtures of B plus other components.

The direction of component B (Table 1) was calculated by regression using an average of seven demagnetization steps (400 to 590°C) yielding average maximum angular deviation (MAD) angles [Kirschvink, 1980] of 2.6° (range 0.8° to 6.1°). The sharp, high temperature unblocking at 560 to 580°C indicates that magnetite is the carrier; in a few specimens a similar direction persists above 590°C (e.g. Figure 2A), suggesting that hematite is also a carrier. It is concluded that the B component in the Matagua is primary, originating in the Aptian or Albian stages (ca. 114 to 95 Ma).

For each site there is an elongate distribution of specimen mean B directions (Figure 3) indicating the probable presence of other magnetizations. Incomplete removal of component A is especially evident in sites CU87.5 and CU89.9. Additional components with intermediate unblocking temperatures, not parallel to A or B directions (e.g. site CU89.8), are most likely associated with burial metamorphism.

Provinciál Formation

Fine grained micritic limestones of the Provinciál (sites CU89.3 and CU89.5) had NRM intensities less than 10^{-4} A/m, and most specimens were unstable during demagnetization and were excluded from further analysis. Volcaniclastic rich samples from sites CU87.4 and CU89.7 (NRM 10^{-2} and 10^{-4} A/m) yielded multicomponent magnetizations that are directionally similar to those of the Mataguá (Figure2B).

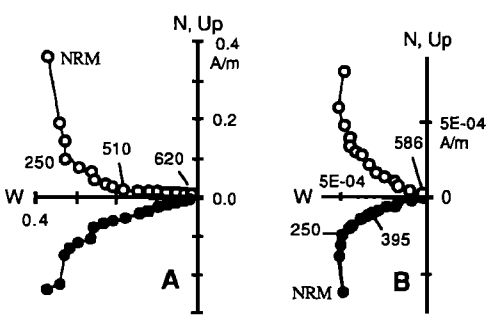


Fig. 2. Representative orthogonal demagnetization diagrams for specimens (A) CU87.05.04 - Mataguá Fm.(B) CU89.07.05 - Provinciál Fm. Open (solid) circles are projections onto horizontal (vertical) plane.

TABLE 1. Paleomagnetic Data Summary

Site	N/N_0	Uncorrected Dec	Uncorrected Inc	k	α_{95}	Corrected Dec	Corrected Inc
*Unscreened Direction**							
87.4(G)	07/09	288.8	18.8	42	9.5	310.5	30.2
89.7(D)	12/16	316.4	28.1	53	6.0	337.0	21.2
87.5(C)	11/11	282.3	27.4	36	7.7	299.7	16.7
89.8(B)	11/12	310.0	36.7	36	7.7	294.1	28.0
89.9(A)	11/13	271.7	31.5	44	7.0	315.4	24.6
Mean	05/05	293.6	29.5	21	16.9		
	05/05			25	15.5	311.4	24.9
	04/05**	287.8	29.3	27	18.0		
	04/05**			59	12.1	304.9	25.1
*Screened Direction**							
87.4(G)	05/09	282.9	17.3	75	8.9	304.8	33.5
87.5(C)	07/11	278.1	20.6	66	7.5	291.9	14.4
89.8(B)	06/12	298.8	41.0	518	4.0	283.1	27.7
89.9(A)	05/13	265.2	22.3	193	5.5	304.1	28.7
Mean	04/05	280.4	25.7	26	18.5		
	04/05			43	14.1	295.7	26.4
*Common axle Direction**							
All	42/61					297.9	26.2

(*) See text. (**) Site CU89.07 omitted. k = (N-1)/(N-R). N/N_0 is ratio of specimens used to those analyzed. Locations (in parentheses after site number) are keyed to Figure 1B and are in sequence from A (oldest) to G (youngest). Structural attitudes (strike/dip) are: (A) 289/70N; (B) 321/26S; (C) 277/42N; (D) 318/43N; (G) 313/46N).

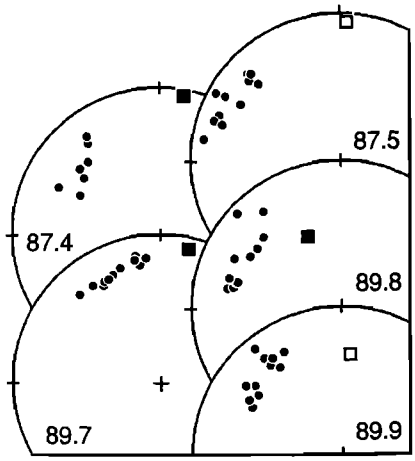


Fig. 3. Equal area projections of higher temperature (B component) directions from each site, in stratigraphic coordinates. Solid (open) symbols indicate downward (upward) inclinations. Squares are structurally corrected Quaternary axial dipole field directions.

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The Provinciál B component (Table 1) differs from the Mataguá B component in having lower and more distributed unblocking temperature spectra. Regression using an average of nine steps (250 to 528°C) had an average MAD angle of 5.2° (range 2.1° to 13.7°)

The specimen mean B directions at each site (Figure 3) are distributed along a great circle similar to the remagnetization circles defined within individual specimens, suggesting again that some of the calculated B directions contain secondary components which overlap the B unblocking temperature spectra. Inability to resolve B from secondary components is most pronounced in site CU89.7 (e.g. Figure 2B), and the site mean "direction" (Table 1) is inaccurate.

Combined Results and Great Circle Analysis

The directional similarity of B components from the two formations, and the similarity of depositional ages, permits the pooling of results. Specimen mean directions within each site are streaked between the B component and A and/or other secondary components. Objective elimination of contaminated specimens was attempted by successively rejecting specimens along each sites' streak from N to W until the remaining specimen directions were entirely within 99% confidence limits (α_{99}) from the iteratively recalculated "screened" site mean direction. The combined "screened" results for four sites (site 89.7 did not converge and was excluded) yield D=295.7°, I=26.4°, k=43.2, α_{95}=14.1° (Table 1). A 1.7-fold increase in k upon structural correction indicates a qualitatively positive fold test. The low increase in clustering results from the similar structural attitudes of most sites and the close parallelism between tilt axes and the B direction.

The methods of Onstott [1980] were used to calculate the direction of intersecting remagnetization circles. The "axles" (unit vectors orthogonal to remagnetization circles) characterizing B plus a secondary component were calculated from demagnetization data between 250 and 590°C for 42 of the 61 total specimens. The mutual intersection point, or common axle (Figure 4), is D = 297.9°, I = 26.2° (stratigraphic coordinates), with a 95% confidence ellipse of 3.8° by 8.6°, azimuth 005°. This direction is within 3° of the screened mean direction. The corresponding (preferred) pole is at 30.7°N Lat., 193.3°E Long. Evaluation of the confidence region for this pole is not straightforward as it must be derived from an elliptical uncertainty interval for the direction which is not symmetric about the declination.

Tectonic implications

The mid-Cretaceous Zaza pole is discordant with the North American Cretaceous APWP (Figure 5). The most appropriate North American mid-Cretaceous reference pole for comparison is that compiled by Globerman and Irving [1988] for the period 118 to 88 Ma, a time interval that encompasses little APWP motion for North America. Discordance between the Zaza pole (28.9°N, 195.0°E, N=4, K=57, A_{95}=12.3°) calculated from screened directions and the contemporary North American pole (71°N, 196°E, N=4, K=352, A_{95}=4.9°) implies an anticlockwise rotation of 43° ± 16° and northward displacement of 7° ± 8° of the sampling region with respect to the North American craton (95% confidence limits calculated after M. Debiche and G. S. Watson, personal communication, 1990).

The inclination calculated by great circle analysis has a 95% confidence interval of about 5°, thus the mid-Cretaceous paleolatitude for the Zaza terrane is 14°± 3°. The expected paleolatitude (at 118-88 Ma) for Zaza sites if attached to the North American plate is 22° ± 5°; accordingly the Zaza terrane was 8°± 6° south of the North American plate in the mid-Cretaceous. Though of limited statistical significance, this

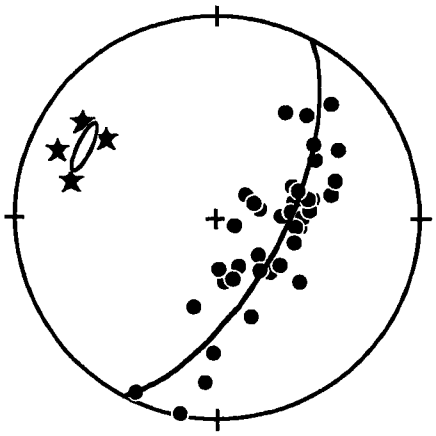


Fig. 4. Intersection direction (common axle) of remagnetization circles, with 95% confidence ellipse. Solid dots are individual specimen axles, shown with best fit great circle. Stars are screened site mean directions for comparison.

paleolatitude discrepancy supports a Caribbean Plate origin for the Zaza terrane. The maximum of 1540 km post mid-Cretaceous northward transport implied for the Caribbean plate supports tectonic models [e.g. Pindell and Dewey, 1982; Duncan and Hargraves, 1984] calling for moderate poleward displacement of the Caribbean plate after its genesis in the Middle Jurassic.

Discussion and Conclusions

The relative rotation implied by discordance between the Zaza pole and North American reference poles is within the angular range of rotations reported for terranes of Jamaica,



Fig. 5. Zaza pole compared with mid-Cretaceous North American reference pole (118 to 88 Ma). North American apparent polar wander path (NA-APWP) from 140 Ma to present. Star denotes sampling area. Zaza pole with A_{95} corresponding to screened mean direction. Stippled band depicts rotation of Zaza pole; its width reflects 95% confidence interval for paleolatitude discordance (8±6°) indicated by common axle uncertainty.

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Hispañola, and Puerto Rico [Van Fossen et al., 1988; 1989; references summarized by Gose, 1985]. For these other terranes, rotation appears to be a consequence of Neogene tectonism in a zone of distributed sinistral shear in the post-collisional transform system. In contrast, central Cuba has been tectonically stable since the Middle Eocene [e.g. Hatten et al., 1989]; presumably rotation of the Zaza terrane occurred earlier than this time, prior to development of the Swan-Oriente transform system. It is presently unclear whether the other Greater Antilles experienced pre-collision rotation.

Pre-transform rotation and northward displacement of the Zaza terrane attests to mobility of the terrane with respect to cratonic North America. Some possible explanations follow:

1) The Zaza terrane represents an island arc originally coupled rigidly to neither the North American nor Caribbean plates, which was sandwiched into the collision zone and accreted to the North American plate. Genetic relationship between the Zaza arc and those of the other Greater Antilles is neither precluded nor expected in this scenario.

2) The Zaza terrane (and possibly all of Cuba) was part of the Cretaceous Greater Antilles arc and originated on the Caribbean plate, which was transported moderately northward [Pindell and Dewey, 1982] and emplaced between the rifted North and South American plates, until the arc collided with the Florida-Bahama platform (Early-Middle Eocene). Rotation may pre-date collision, in which case the arc was originally more north-south in orientation. Alternatively, the rotation may have been syn-collisional or even slightly post-collisional, possibly a consequence of transpression.

In either case, the existence of a south-dipping subduction zone north of the Zaza terrane is suggested. A Caribbean plate transport history for Cuba is consistent with evidence from basement tectonothermal history in the Las Villas terrane, north of the Las Villas thrust [Renne et al., 1989].

Two unresolved questions are central to further elucidation of the tectonic history of the northern Caribbean margin. First, the age of Zaza rotation must be better established through paleomagnetic research on rock units younger than mid-Cretaceous. Second, it must be resolved whether other arc terranes of the Greater Antilles experienced pre- Middle Eocene rotation consistent with that seen in the Zaza terrane. Whether these terranes have a common origin and subsequent accretion history remains to be seen.

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